

Bairui Zhang

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RESEARCH INTERESTS

Reliable agentic reasoning; multimodal and embodied agents; tool use, memory, and self-evolution; knowledge-grounded reasoning; rigorous and resource-aware evaluation of LLM agents.

EDUCATION

- The Hong Kong University of Science and Technology (Guangzhou)** Sep. 2025 – Jun. 2027 (expected)
M.Phil. in Data Science and Analytics; Full Research Scholarship (RMB 240,000) Guangzhou, China
- GPA: **4.0/4.0**; all completed courses graded A. KiMI Lab; advised by Prof. Yongqi Zhang and Prof. Sihong Xie, with Prof. Lei Chen (IEEE Fellow) as lab mentor.
- Northeastern University** Sep. 2021 – Jun. 2025
B.E. in Computer Science and Technology Shenyang, China
- GPA: **3.99/4.0** (92.4/100); ranked **3/116** in the major (top 2.5%); average above 95 in core courses. Outstanding undergraduate thesis on tree-search-based fine-grained image understanding for LLMs.

SELECTED RESEARCH EXPERIENCE

- VeriGraph: Structured Knowledge-Graph Fact Verification** May 2025 – Sep. 2026
First Author
- Developed a verification pipeline that decomposes natural-language claims into entity–relation–value–time constraints and introduced a **Shared-Witness** mechanism to prevent invalid evidence composition across different entities.
 - Achieved **84.07% accuracy / 81.68% F1** on all 9,041 official FactKG test cases, improving over the Joint-Graph baseline by **7.33 / 10.97 percentage points**; corrected 672 cases with only 9 regressions.
 - Designed a four-state knowledge-maintenance policy (supported, contradicted, missing, uncertain) that invokes web evidence only for missing knowledge, reducing external searches from 5,579 to 2,072 (**–62.86%**); manuscript in preparation.
- DGSE: Reusable Skills from Multimodal-Agent Failures** Dec. 2025 – May 2026
Project Lead and First Author; Huawei Open Challenge
- Built a closed-loop perception–reasoning–execution agent with retrieval, code-execution, and visual-operation tools; curated **1.08M** SFT examples from 20+ task sources through hard-case mining and augmentation, improving related benchmarks by **5–15%**.
 - Proposed DGSE, which converts failed trajectories on structured visual tasks into executable Python skills through visual program synthesis and diagnostic repair.
 - Improved three benchmarks, including TIR-Bench and PuzzleVQA, by **12.1 points on average**; deployed skills ran **100× faster**, with zero LLM calls and sub-second execution, and transferred across models.
- TokenCliff: Budget Elasticity in Multi-Turn LLM Agents** Feb. 2026 – May 2026
First Author; EMNLP 2026 Main Conference
- Built 660 deterministic tasks across five domains, with seeded programs generating and verifying environments and answers without LLM judges; evaluated 10 models at five token budgets (200–8,192), totaling **33,000 episodes**.
 - Introduced **Budget Elasticity** and **Cliff Index**; identified adjacent-budget accuracy drops of up to **32 points** and found that neither peak performance nor model scale reliably predicts budget robustness.

ADDITIONAL RESEARCH EXPERIENCE

- Sepsis Mortality and Personalized Medication Modeling** Jul. 2023 – Oct. 2023
Summer Research Project; Advisor: Prof. Manolis Kellis, Massachusetts Institute of Technology
- Developed mortality-risk and personalized-medication models using SVM, Random Forest, and XGBoost; achieved **73% accuracy**, received **98/100**, and ranked first in the project group.
- Artificial Intelligence in Pancreatic Image Analysis: A Review** Mar. 2023 – Jul. 2024
Undergraduate Researcher; Second Author
- Synthesized AI methods for classification, segmentation, detection, and prognosis across CT, MRI, EUS, PET, and pathology in a 22,000-word review published in *Sensors*, 24(14):4749.

PUBLICATIONS AND MANUSCRIPTS

- [1] **Bairui Zhang**, Weixuan Liu, Defan Xue, and Yongqi Zhang. “TokenCliff: Benchmarking Budget Elasticity in Multi-Turn LLM Agents.” *EMNLP 2026 Main Conference*. **First author**.
- [2] Xing Mu, Xiaotian Lin, Zhangyang Peng, **Bairui Zhang**, and Yuyu Luo. “From Global to Regional Coverage: Hyperspherical Prototype Discovery for Storage-Efficient Multimodal Retrieval.” *EMNLP 2026 Main Conference*.
- [3] Zirui Song, Yuan Huang, Junchang Liu, Haozhe Luo, Chenxi Wang, Lang Gao, Zixiang Xu, **Bairui Zhang**, Mingfei Han, Xiaojun Chang, and Xiuying Chen. “Beyond Survival: Evaluating LLMs in Social Deduction Games with Human-Aligned Strategies.” *EMNLP 2026 Main Conference*.
- [4] Weixuan Liu, **Bairui Zhang**, Tao Liu, Juntao Jiang, and Yong Liu. “Artificial Intelligence in Pancreatic Image Analysis: A Review.” *Sensors*, 24(14):4749, 2024. doi:10.3390/s24144749.
- [5] Xiaoxiao Liu, Qingying Xiao, Junying Chen, Xiangyi Feng, Xiangbo Wu, **Bairui Zhang**, Xiang Wan, Jian Chang, Guangjun Yu, Yan Hu, and Benyou Wang. “Large Language Models for Outpatient Referral: Problem Definition, Benchmarking and Challenges.” *arXiv preprint*.
- [6] **Bairui Zhang**, Hange Zhou, Weixuan Liu, Hongyu Ge, Xing Mu, and Yongqi Zhang. “DGSE: Turning Agent Failures into Reusable Skills for Structurally Regular VQA.” Under review at *ICLR 2027*. **First author**.
- [7] Pengxu Zhu, Weixuan Chen, Yang Liu, **Bairui Zhang**, Qishi Zhan, and Yiwei Wang. “Information Asymmetry as a Design Axis for Step-Level Process Verification.” Under review at *AAAI 2027*.
- [8] Defan Xue, Jiabao Yang, **Bairui Zhang**, Guangyu Hu, Qixiang Yin, and Song Xiao. “WFCLLM: Learning Where and How to Embed Semantic Watermarks in LLM-Generated Code.” Under review at *AAAI 2027*.
- [9] Yuhang Wu, Qinyuan Liu, Qiuyang Zhao, Bo Jiang, **Bairui Zhang**, Yu-Jie Xiong, Hao Zhang, and Qingwei Chong. “Freeze Deep, Train Shallow: A Framework for Layer Allocation in Continued Pre-Training.” Under review at *ACL Rolling Review*, Aug. 2026.
- [10] Defan Xue, Chenyu Wu, **Bairui Zhang**, Jiabao Yang, and Song Xiao. “When Do LLM Watermarks Lose Information? A Diagnostic Study under Controlled Rewriting.” Under review at *ICLR 2027*.

INDUSTRY EXPERIENCE

Sangfor Technologies, Innovation Research Institute

Aug. 2026 – Present

Algorithm Engineer, X-Star Talent Program

Shenzhen, China

- Developing an auditable multi-agent workflow for vulnerability research in large codebases, with bounded exploration over competing security hypotheses and reproducible vulnerability-exposure scripts as verification evidence.
- Representing verified facts, counterevidence, and failed attempts as shared evidence state; dynamically allocating compute to branches with the highest expected information gain to reduce anchoring and duplicated search.

Ericsson

May 2026 – Aug. 2026

Agent Algorithm Engineer Intern

Guangzhou, China

- Improved BM25/PageIndex hit rate by approximately **10%** for 5G Core fault diagnosis through domain-term extraction and query normalization across abbreviations, aliases, synonyms, and mixed Chinese–English queries.
- Designed an eight-dimensional diagnostic rubric and LLM-as-Judge rules; built a 137-field auto-fill tool that reduced manual report completion from 30+ to about 12 minutes per case and was adopted in the team’s standard workflow.
- Contributed to a multi-agent harness with task routing, quality gates, state machines, DAG execution, and structured memory; received an **A+ internship evaluation**.

SELECTED HONORS AND AWARDS

- Meritorious Winner, Mathematical Contest in Modeling, 2024; First Prize, Asia and Pacific Mathematical Contest in Modeling, 2023.
- Ranked first in Prof. Manolis Kellis’s MIT summer research group and received a recommendation letter, 2023; fully funded Singtel visit, 2025; invited participant, Future Science Prize Week, 2024.
- Named inventor on three national patents, 2023; First-Class University Scholarship, 2024; Outstanding Graduate, 2025; University Merit Student, 2021–2025.

TECHNICAL SKILLS

Agents and LLMs: LangChain, LangGraph; reasoning, tool use, memory, multi-agent orchestration; RAG, BM25, LLM Wiki; SFT, LoRA, benchmark design, checklist-based evaluation.

Training and Systems: PyTorch, MS-Swift, vLLM, SGLang; Python, C++, SQL (Hive/Spark), MATLAB; Linux, Docker, Kubernetes, Git.

Languages: Mandarin Chinese (native); English (IELTS 6.5, CET-6 540).